

Date :-

Expt No: 03

ENCODER AND DECODER

ENCODER:-

AIM:-

To construct and verify 9-to-4 encoder using IC 74147.

APPARATUS REQUIRED:-

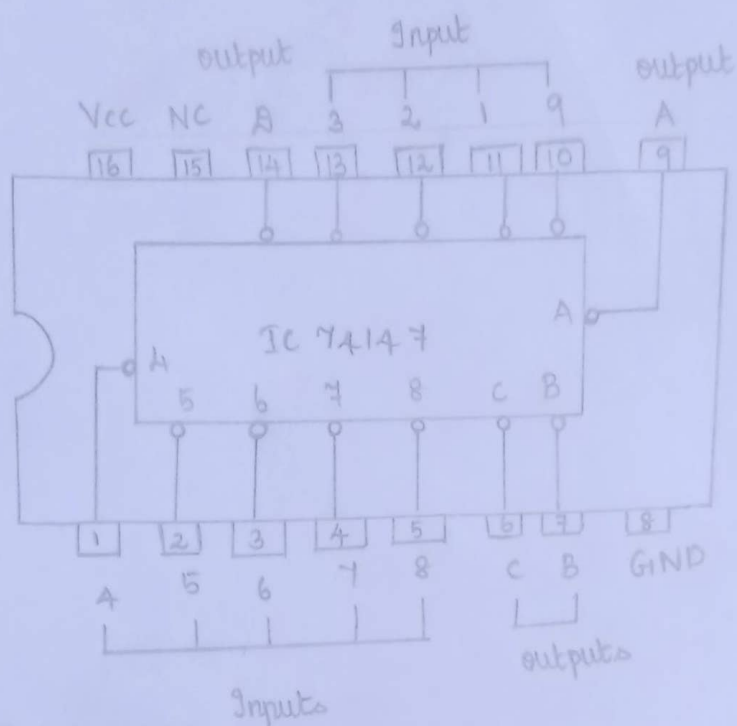
- 1) 9 to -4 Encoder - IC 74147
- 2) Connecting wire
- 3) Digital trainer kit / breadboard
- 4) +5V DC supply.

THEORY:-

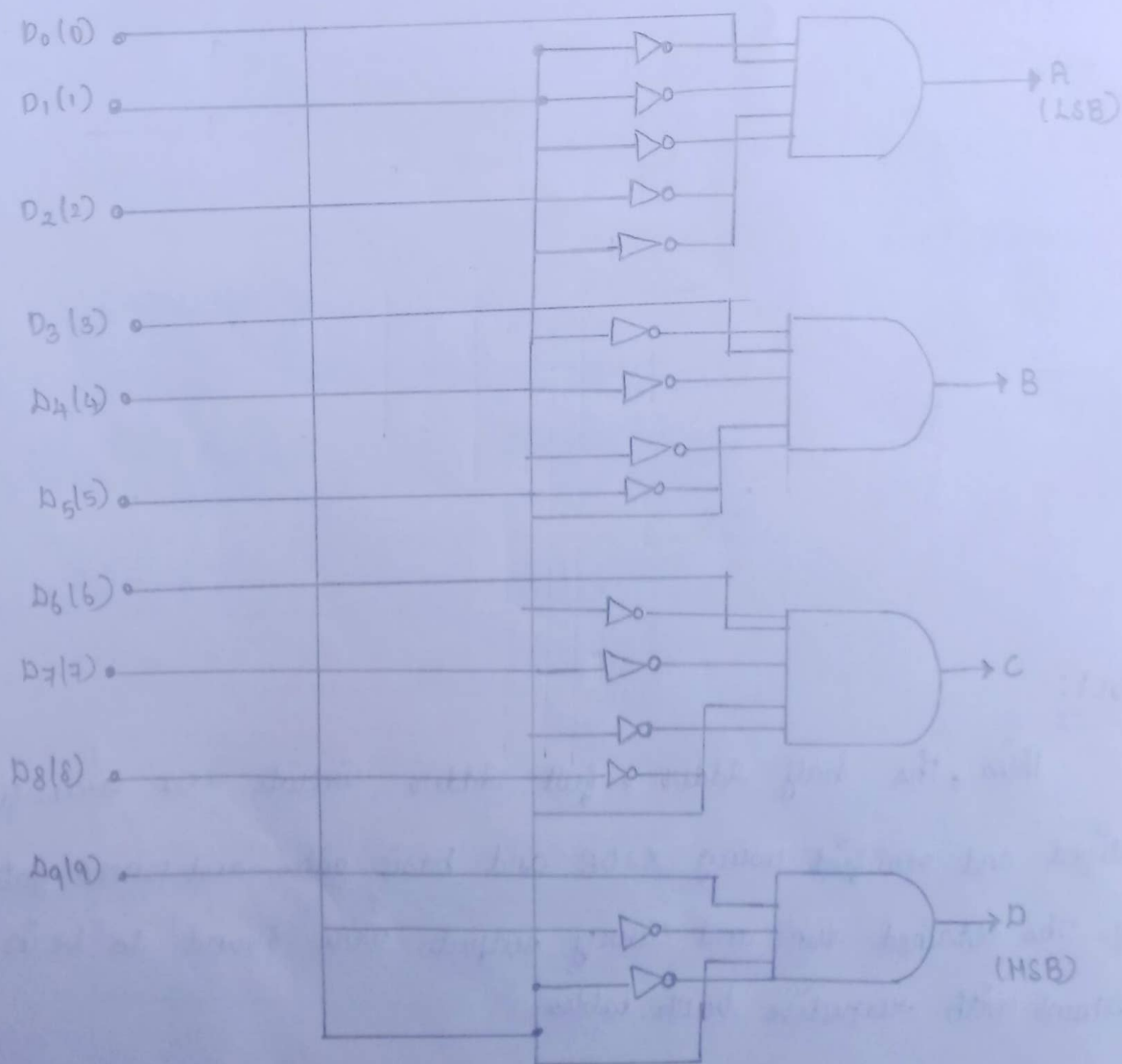
An encoder is a combinational circuit that converts information from a larger number of input lines into a smaller number of output lines.

A 9-to-4 Encoder has 9 input lines and 4 output lines. The inputs represent decimal digits 0 to 9 and the corresponding BCD code at its outputs. Since the inputs and outputs of IC 74147 are active low, the selected input is connected to LOW and the corresponding outputs are obtained in inverted form.

IC 74147 Pin Diagram:-



CIRCUIT DIAGRAM:-



PROCEDURE :-

- 1) Place the IC 74147 on the digital trainer kit / breadboard
- 2) connected the IC to the required +5V supply and ground.
- 3) Connect the input switches to the input pins of the IC
- 4) Give the LOW to the required input corresponding to the decimal digit.
- 5) observe the four output lines
- 6) Record the output for each input condition
- 7) Verify the obtained output with the function table.

RESULT :-

Thus, the 9 to 4 Encoder was constructed and verified using IC 74147

DECODER:-

AIM:- To construct and verify 3×8 decoder using IC 74138

APPARATUS REQUIRED:-

(i) 3×8 decoder - IC 74138

(ii) Connecting wires.

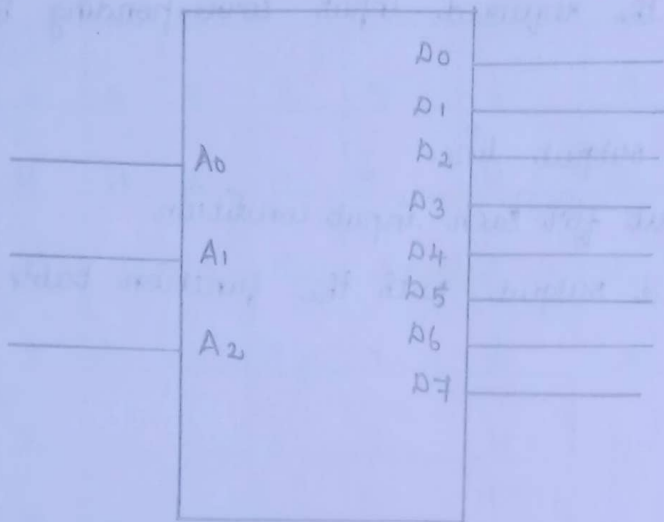
THEORY:-

Discrete quantities of information are represented in digital systems by binary codes. A binary code of n bit can represent n th power of 2 information. A decoder is a combinational circuit that converts binary information from n input lines. These decoders are called n to m line decoders. Where m may be lesser than or equal to the n th power of 2. Their purpose is to generate n th power of 2 minterms of n input variables. The name decoder is also used in conjunction with other code converters; an eg. is BCD to seven segment decoder. The 3 inputs are decoded into 8 outputs. Each representing one of the minterms of 3 input variables. A particular application of this decoder is octal to binary conversion. The input variables represent a binary number and the output represents 8 digit number in octal number system.

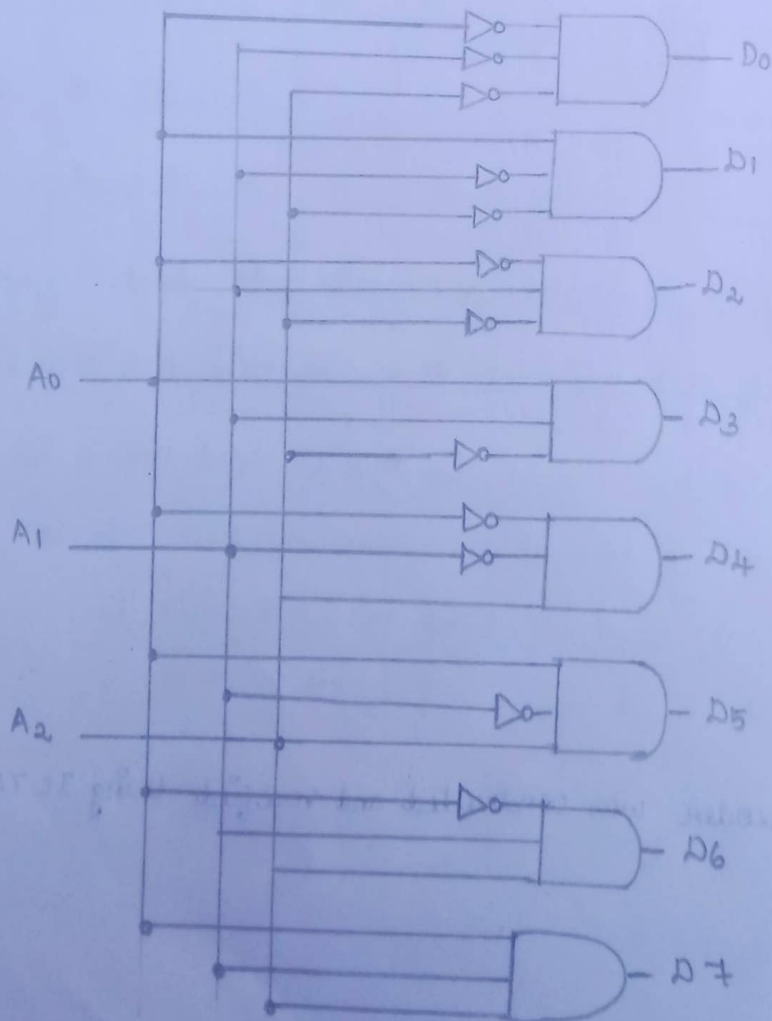
PROCEDURE:-

The 3×8 decoder is mainly used for binary to octal conversion. The circuit connections are given as shown in the circuit diagram. Inputs are given to the 3 invertors. The each of the AND gates generates one of the minterms for the inputs. Then the decimal decoded output is verified.

PIN Diagram for (IC 74138)



CIRCUIT DIAGRAM:-



[illegible]



Input A	Input B	Input C	Output Y0	Output Y1	Output Y2	Output Y3	Output Y4	Output Y5	Output Y6	Output Y7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

RESULT:-

Thus, the 3x8 decoder is constructed and verified using the digital IC 74138.